

Sinhala/Tamil Speaker Verification — Progress Report

Benchmark v1 complete: 54/54 trained, 54/54 scored on held-out test

SL-SPV project

2026-08-22

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1. Where this sits, and what changed since 17 August

This continues the programme recorded in `2026-08-17-progress-report-architecture-frontend-and-corpus-study.md`, which is **not superseded and has not been altered** — it remains the record of what was known on 17 August. This report states what changed.

Phase	Date	Established
P0–P3	2026-07-03	Benchmark v0 , closed-set, SLR52 + SLR65. Thesis ch. 1–4 and <code>papers/ieee_spl</code> written against it.
Dataset build + QC	2026-08-10	Eight corpora unified; two speaker-identity bugs found; Tamil pool 50 -> 752 speakers.
Benchmark v1 (interim)	2026-08-12–17	Staged study launched. 50/54 trained, 24/54 scored . All headline numbers were <i>validation</i> EER.
Benchmark v1 (final)	2026-08-18–22	54/54 trained, 54/54 scored on the held-out test set, every rebuild verified against its checkpoint.

The 17 August report carried an explicit caveat: “*All figures are validation EER on single runs. They become claims only after the held-out evaluation and the paired bootstrap complete.*” That condition is now met.

1.1 The four things that had to be fixed first

Completing the benchmark was not a matter of waiting. Four defects stood between the 17 August state and a trustworthy table, and three of them would have produced **confident, plausible, wrong numbers** rather than an error.

#	Defect	Consequence had it stood
1	Validation dataloader deadlock in the two <code>ssl_wavlm_ecapa</code> runs	Both had been stuck since 16 August — 6 attempts, always at the first validation after a resume. Stage H would have been reported as 2/4.
2	<code>si_pooled</code> keyed on condition <i>name</i> , not on data	KeyError ; all seven pooled-Sinhala runs unevaluable, losing the read-vs-wild contrast entirely.
3	<code>--ssl_*</code> arguments absent from <code>resolved_parameters</code>	Evaluation rebuilt SSL models from model defaults : <code>ssl_layer</code> fell back to -1. <code>F_ssl_wavlm_low</code> , trained on layer 0, would have been <i>scored on layer 12</i> — every tensor matching, no warning.

#	Defect	Consequence had it stood
4	Trainer argparse defaults not applied at rebuild	<code>--encoder_type</code> defaults to SAP in the trainer and ASP in the model class. SSL models were rebuilt with ASP (1536-dim) against SAP (768-dim) checkpoints: 229 encoder tensors loaded, bn.* and fc.weight did not, leaving the final projection at random initialisation . Nine evaluations reported EERs from partly-random weights.

Defects 3 and 4 share one root cause and it is worth naming precisely: **the trainer passes its entire parsed namespace to the model, so a flag nobody mentions still reaches the model carrying the trainer’s default**. Any evaluator that rebuilds from a manifest instead of from that namespace inherits the *model’s* defaults instead, and the divergence is silent whenever it does not change a tensor shape.

Fixes, in `experiments/tools/evaluate.py`:

- `argv_parameters()` recovers arguments from `command.json`, the literal `argv`.
- `trainer_defaults()` AST-parses `trainSpeakerNet.py` for all 82 argparse defaults — closing the whole class rather than the two flags observed.
- A skipped backbone tensor is now a **hard failure**, not a warning. It should never have been a warning: a wrong number is worse than no number.
- A prior `test_eval.json` is archived to `eval_history/` before any overwrite.

All 54 experiments now rebuild with zero backbone mismatch.

2. Headline results (held-out test, speaker-disjoint)

2.1 Best system per language

Language	System	Cosine EER	AS-Norm EER	minDCF
Sinhala	<code>ssl_mhubert_lw</code> (layer-weighted mHuBERT-147)	2.369	2.178	0.156
Tamil	<code>ssl_mhubert_ecapa</code> (mHuBERT + ECAPA head)	0.402	0.321	0.026
Tamil (cosine)	<code>ssl_wavlm_ecapa</code>	0.392	0.392	0.037

Against the strongest mel-filterbank baseline (`ecapa1024`: 4.375 si / 0.884 ta), the layer-weighted SSL front end is **46 % better on Sinhala** and the SSL+ECAPA hybrid **55 % better on Tamil**.

2.2 The layer result, now on held-out test

The 17 August report predicted a mid-stack peak and was falsified. The held-out test set confirms the falsification and sharpens it:

Front end	si EER	ta EER
layer 0 (lowest)	4.033	1.125
layer 6 (mid)	5.525	1.898
last layer (the conventional choice)	8.247	4.620
13 learned layer weights	3.297	0.623

Speaker information falls **monotonically with depth**, and the conventional `--ssl_layer -1` is the worst available choice. Learned weighting beats every single layer, so the gain is not merely “use a low layer” — the mixture carries information no individual layer does.

2.3 A new result: the last-layer penalty scales with linguistic distance

Reading the same frozen WavLM at its last layer costs almost nothing in English and is catastrophic in Tamil:

Language	SSL last layer	mel ECAPA-1024	penalty
English (<code>en_matched</code>)	7.880	7.580	1.04x
Sinhala	8.247	4.375	1.88x
Tamil	4.620	0.884	5.23x

WavLM is pretrained on English. Its upper layers specialise toward English phonetic content, so discarding the lower layers removes speaker information that the English task can partly recover from context and the Sri Lankan languages cannot. **This is the clearest evidence in the study that last-layer SSL practice is an English-centric default, and that its cost is a function of the target language.** It also explains why the English control was worth running: without it this would have looked like an SSL failure rather than a transfer failure.

3. Domain, channel and corpus effects

3.1 Read speech versus in-the-wild Sinhala

The `si_pooled` condition trains on read + in-the-wild Sinhala and is scored separately on each:

Backbone	read	wild	ratio
ecapa1024	3.297	16.501	5.0x
ecapa512	3.357	16.978	5.1x
resnetse34v2	3.811	18.887	5.0x
mlpmixer	4.275	17.638	4.1x
vggvox	5.384	23.660	4.4x
resnetse34l	5.212	21.527	4.1x
ssl_wavlm	7.965	27.508	3.5x

Genuine cross-session audio is **4–5x harder** than read speech for every backbone. Trained on in-the-wild Sinhala alone (`si_celeb`, 82 speakers) the task is harder still — 19.7 % to 27.3 % EER. Any deployment claim resting on read-speech numbers is overstated by roughly a factor of five.

3.2 Tamil’s advantage is partly, but not wholly, a channel artefact

SLR127 pools three collection sites; a same-batch-only impostor list removes the channel cue:

System	pooled list	same-batch	inflation
ecapa1024	0.884	1.301	1.47x
ssl_mhubert_lw	0.422	0.562	1.33x
ssl_mhubert_ecapa	0.402	0.505	1.26x
ssl_wavlm_ecapa	0.392	0.563	1.44x

The effect is consistent (1.26–1.47x) and must be reported, but Tamil remains substantially easier than Sinhala even after control: 1.30 % against 4.38 % for the same backbone. The asymmetry is real; its *magnitude* was inflated.

3.3 English control: the architecture ranking does not transfer

Backbone	English en_matched	Sinhala	Tamil
ecapa1024	7.580	4.375	0.884
resnetse34v2	7.260	4.708	1.376
ssl_wavlm (last)	7.880	8.247	4.620
vggvox	10.420	5.847	1.949

On English, `resnetse34v2` edges `ecapa1024` and last-layer SSL is competitive. On Sinhala and Tamil, `ecapa1024` leads and last-layer SSL collapses. **An architecture ranking established on English does not transfer to these languages**, which is the direct answer to the question that motivated the English arm.

4. Status of the proposed architectures (A1–A9)

`proposals/` implements nine architectures from the literature study, as self-contained code that never modifies `voxceleb_trainer/`. `selftest.py` reports **13 passed, 0 failed**, including the isolation assertion.

Implementation is not evidence, and the two must not be conflated:

	Implemented	Runner	Run on real data
A2 CC-NAP	yes	yes	started 22 Aug
A3 SL-AMEC	yes	yes	yes (14 result files)
A1, A4, A5, A6, A7, A8, A9	yes	no	no

So **one of nine** has real results, a second is now running, and seven have never been executed against a trained model. The plan for those is section 6.

5. What is now safe to claim, and what is not

Safe. Every number in sections 2 and 3 is held-out test EER on speaker-disjoint splits, from a model verified to match its own checkpoint.

Not yet safe. Single-seed results. The power analysis in the 17 August report still binds: effective sample size is capped by *speaker* count, not trial count ($n_{\text{eff}} \rightarrow S/\rho$), so a single absolute EER carries roughly ± 5 –6 pp at $S = 91$. Differences of the size seen between `ecapa1024` and `ecapa512` (0.09 pp) are **not** resolvable; differences of the size seen between `ssl_mhubert_lw` and `ecapa1024` (2.0 pp on Sinhala) are. The paired speaker-clustered bootstrap over identical trials is the correct instrument and is running.

Explicitly still open. The §4.2 arbiter from the 17 August report — layer-weighted *and* fine-tuned WavLM on open-set data — remains unrun. v1 measured layer-weighted+frozen (2.369 si) and single-layer+fine-tuned (5.404 si) but never the diagonal that P3’s headline claim actually rests on.

6. Next steps, in priority order

1. **The arbiter run.** Layer-weighted + fine-tuned, open-set. One run per language. Directly tests P3’s “full fine-tuning beats LoRA” claim under v1’s protocol; that claim currently underpins `papers/ieee_spl`.
2. **A7 PLLW** (per-language layer weighting). The cheapest high-value proposal and the direct extension of this report’s strongest result: if the optimal layer differs between Sinhala and Tamil, one shared weight vector is leaving performance on the table. ~39 extra parameters.
3. **A9 ARI-SubCenter.** A loss-only change, so it reuses the existing recipe unchanged. Motivated directly by the QC audit: sub-centres absorb the label noise the audit proved is present.
4. **Seeds.** Three seeds for the top three systems per language, to convert single-run numbers into intervals.
5. **A2/A3 completion**, then the trainable proposals A1/A4/A5/A6/A8.

7. Reproduction

status of everything, any time

```
python3 experiments/tools/status.py
```

```
python3 experiments/tools/status.py --watch 60
```

outstanding evaluations spread over every free GPU

```
python3 experiments/tools/eval_dispatch.py --run
```

aggregate + paired bootstrap

```
python3 experiments/tools/analyze.py --n-boot 2000 --out experiments/analysis/v1-final
```

corpus copy to the dedicated ricproject5 spindle (waits for an idle cluster)

```
python3 experiments/tools/copy_to_ricproject5.py
```

Prior evaluations are preserved in `experiments/results/*/eval_history/` and in `experiments/results_backup/pre-redo-`
No previous report has been modified or removed.